DATABASE DESIGN

Planning the Database

From requirements to a conceptual model

Requirements / Entities / Attributes

Relationships / Cardinality / ER model

Practical Web Database Design

Chapter 5: Planning the Database

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02 Start with the business

CUSTOMER

- Browse products
- Place an order
- Track order status

STORE OWNER

- Manage product inventory
- Fulfill and ship orders
- Analyze business performance

The database must support the required business behavior.

03 The four planning steps

01 Gather information

- 02 Identify entities and attributes

03 Identify relationships

- 04 Prepare the conceptual model

Requirements lead to a model before implementation.

04 Gather information deliberately

ASK STAKEHOLDERS

- What must customers do?
- What must the owner manage?
- What questions must reports answer?

LOOK FOR STORED DATA

- Product information
- Order and customer information
- Rules and exceptional cases

A plausible guess can still be wrong for this project.

05 Product information

DESCRIBE AND DISPLAY

- Name; short and long descriptions
- Size; color; weight
- Picture or image
- Comments from satisfied customers

SELL AND FULFILL

- Price
- Availability / quantity on hand
- Special discounts
- Shipping cost

This is an information inventory, not a finished table.

06 Administration reveals requirements

OWNER OPERATIONS

- Add new products
- Change descriptions and prices
- Move products between categories
- Offer special discounts

ACCESS CONTROL

- Recognize the store owner
- Authorize administrative functions
- Keep administration restricted
- Account details excluded from this model

Processing requirements can reveal additional data to store.

07 Remove products; preserve history

BEFORE

- A product has existing sales
- Historical orders refer to it
- The owner stops selling it

AFTER

- Keep the product record
- Set status to deleted
- Hide it from customer listings

Soft deletion removes visibility while retaining historical references.

08 Reports affect database planning

BUSINESS QUESTIONS

- Which products sell most or least?
- Which categories perform best?
- Which customers buy which products?
- Were discounts effective?

DESIGN CHECK

- Can stored data answer each question?
- Can sales be attributed correctly?
- Can discounted purchases be identified?
- What information is missing?

A report cannot recover information the system never captured.

09 Order information has different levels

ORDER AS A WHOLE

- Total shipping cost
- Taxes, if applicable
- Total cost
- Order-level discounts, if required

PRODUCT WITHIN AN ORDER

- Which product was ordered
- Quantity of that product
- Item-level discounts, if required
- A relationship with Product

Separate order facts from facts about each ordered product.

10 Customer information: collect with purpose

NEEDED FOR THE PURCHASE

- Valid payment authorization
- Where to ship the products
- Possibly how to ship them

ADDITIONAL INFORMATION

- Marketing or analysis questions
- Example: gift or personal use?
- Usually optional
- Forced answers can cause abandonment

Distinguish fulfillment needs from optional information.

11 Email and tracking add data needs

EMAIL FOLLOW-UP

- Offer shipment confirmation
- Store an email address
- Support the expected follow-up

RETURNING CUSTOMERS

- Recognize the customer
- Show that customer's orders
- Customer login ID and password

Implementation clarification: store a password hash, not plaintext.

12 Payment processing and payment outcome

THIRD-PARTY PROCESSOR

- Customer enters payment information
- Processor authorizes the payment
- Returns approved or denied

WEB STORE

- Records the outcome on the order
- May allow saving a denied order
- May support another attempt later

A denied payment does not automatically mean deleting the order.

13 Entities, attributes, relationships

ENTITY AND ATTRIBUTE

- Entity: a thing we store data about
- Example: Product
- Attribute: a property of that entity
- Example: Product price

RELATIONSHIP

- An association between entities
- Product belongs to Category
- Customer places Order
- Order includes Product

Separate descriptive properties from associations.

14 The preliminary entity inventory

PRODUCT AND CATEGORY

- Product: descriptions, dimensions, price
- Stock, image, comments, discounts
- Shipping cost; status discovered earlier
- Category: name and description

CUSTOMER AND ORDER

- Customer: name, email, ship-to address
- Login ID and password representation
- Order: shipping, taxes, total, discounts
- Payment outcome discovered earlier

Reconcile the attribute list with every requirement already discovered.

15 Ask in both directions

FIRST DIRECTION

- Each X relates to how many Y?
- Record the minimum
- Record the maximum

REVERSE DIRECTION

- Each Y relates to how many X?
- Ask independently
- Confirm against requirements

One direction does not establish the other.

16 Read minimum and maximum cardinality

OPTIONAL PARTICIPATION

- 0..1 = zero or one
- 0..* = zero or more
- Minimum zero permits no relationship

MANDATORY PARTICIPATION

- 1..1 = exactly one
- 1..* = one or more
- Minimum one requires a relationship

Crow's foot = many; circle = zero; bars indicate one.

17 Category and Product: many-to-many

CATEGORY TO PRODUCTS

- Each category contains 1..* products
- At least one product is required
- Empty categories should not be shown

PRODUCT TO CATEGORIES

- Each product belongs to 1..* categories
- At least one category is required
- Multiple categories are allowed

These minimums are decisions for the example store.

18 Default category: relationship data

SUPPLEMENTARY EXAMPLE

- Sports shirt belongs to Clothing
- The same shirt belongs to Sportswear
- One sale occurs

REPORTING DECISION

- Which category receives sales credit?
- Choose a default category
- Default applies to this product link

The default designation describes Product + Category together.

19 Order and Product: many-to-many

PRODUCT TO ORDERS

- Each product appears in 0..* orders
- A new product may have no orders
- The product may be ordered repeatedly

ORDER TO PRODUCTS

- Each order contains 1..* products
- An order needs at least one product
- Several products may be bought together

A zero minimum does not change the many-to-many maximums.

20 Ordered quantity belongs to the link

SUPPLEMENTARY EXAMPLE

- Order A + Notebook = quantity 2
- Order B + Notebook = quantity 5
- Same product; different order facts

LATER IMPLEMENTATION

- Introduce an associative entity
- Often called OrderItem or OrderDetail
- Store quantity for the order-product pair

Quantity ordered is neither global stock nor an order-wide quantity.

21 Category and Subcategory: one entity

SAME TYPE OF THING

- Both have a name and description
- Subcategory describes a role
- Use Category at every level

RECURSIVE RELATIONSHIP

- Category relates to Category
- One instance acts as parent
- Another acts as child

A hierarchy does not require a new entity for every level.

22 Hierarchy rules and category identity

CARDINALITY

- Category to children: 0..*
- Category to parent: 0..1
- Root has no parent; leaf has no children

IMPORTANT DISTINCTIONS

- One parent is a design decision
- Recursive can be many-to-many elsewhere
- Same name does not mean same category

The example chooses a single-parent hierarchy.

23 Customer and Order: one-to-many

CUSTOMER TO ORDERS

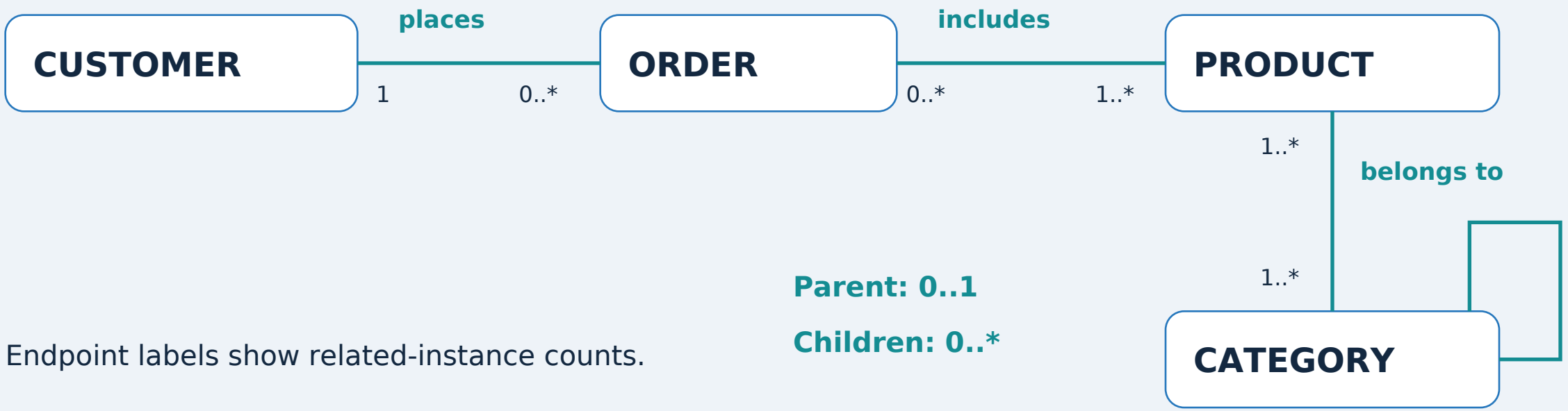
- Each customer has 0..* orders
- Registration may precede a purchase
- Repeat purchases are allowed

ORDER TO CUSTOMER

- Each order has exactly 1 customer
- The application knows who ordered
- Shared customer ownership is excluded

A customer may exist without an order; the reverse is not allowed.

24 Assemble the conceptual model



Endpoint labels show related-instance counts.

Parent: 0..1

Children: 0..*

Keep both direction rules visible when reading the model.

25 From conceptual model to relational design

CONCEPTUAL MODEL

- Entities and their meaning
- Relationships and participation rules
- Business decisions and exceptions

NEXT: IMPLEMENTATION

- Resolve Product-Category association
- Resolve Order-Product association
- Choose keys, types, and constraints

Use an associative entity to represent each many-to-many association.

26 Interview: how do you determine cardinality?

A CLEAR ANSWER

- Start with stakeholder rules
- Ask the question in both directions
- Record minimum and maximum
- Validate with edge cases

SUPPLEMENTARY CHECK

- Registered customer, no orders: valid
- Order without customer: invalid
- New product, no orders: valid
- Order without products: invalid

Explain the business evidence behind each cardinality.

27 Practical project use

SUPPLEMENTARY: DOCUMENT SYSTEM

- Identify documents and their attributes
- Identify users and departments
- Ask who may own or receive documents

APPLY THE SAME METHOD

- Can a document have no owner?
- Can it have multiple recipients?
- Which facts belong to each assignment?

Reuse the questions; confirm the rules for the actual project.

28 Chapter review and key takeaway

REVIEW

- Gather information before modeling
- Separate entities, attributes, relationships
- Ask both direction questions
- Preserve history and reporting needs

NEXT CHAPTER

- Carry the conceptual model forward
- Resolve the two many-to-many links
- Place relationship data correctly
- Design the relational implementation

KEY TAKEAWAY: Model the business rules before designing the tables.

Created with AI assistance. Reviewed and presented by Abdalla Fadl.